

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

1. A land snail, a clam, and an octopus all share
(A) a mantle. (C) gills.
(B) a radula. (D) distinct cephalization.
2. Which phylum is characterized by animals that have a segmented body?
(A) Cnidaria (C) Arthropoda
(B) Platyhelminthes (D) Mollusca
3. The water vascular system of echinoderms
(A) functions as a circulatory system that distributes nutrients to body cells.
(B) functions in locomotion and feeding.
(C) is bilateral in organization, even though the adult animal is not bilaterally symmetrical.
(D) moves water through the animal's body during filter feeding.
4. Which of the following combinations of phylum and description is correct?
(A) Echinodermata—radial symmetry as a larva, coelom
(B) Nematoda—roundworms, internal skeleton
(C) Platyhelminthes—flatworms, gastrovascular cavity, no body cavity
(D) Porifera—gastrovascular cavity, coelom

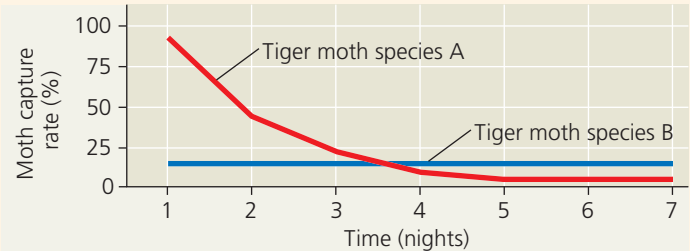
Levels 3-4: Applying/Analyzing

5. In Figure 33.1, which two main clades branch from the most recent common ancestor of the eumetazoans?
(A) Porifera and Cnidaria
(B) Lophotrochozoa and Ecdysozoa
(C) Cnidaria and Bilateria
(D) Deuterostomia and Bilateria
6. **MAKE CONNECTIONS** In Figure 33.7, assume that the two medusae shown at step 4 were produced by one polyp colony. Review Concept 12.1 and Concept 13.3, and then use your understanding of mitosis and meiosis to select which of the following is true.
(A) Both the medusae and the gametes are genetically identical.
(B) Neither the medusae nor the gametes are genetically identical.
(C) The medusae are not genetically identical but the gametes are genetically identical.
(D) The medusae are genetically identical but the gametes differ genetically.

Levels 5-6: Evaluating/Creating

7. **EVOLUTION CONNECTION • INTERPRET THE DATA** Based on Figure 32.11, draw a phylogenetic tree of Bilateria that includes the ten phyla of bilaterians discussed in detail in this chapter. Label each branch that leads to a phylum with a C, H, HC, or N, depending on whether members of the phylum have a coelom only (C), hemocoel only (H), hemocoel and reduced coelom (HC), or no body cavity (N). Use your labeled tree to answer the following questions. (a) For each of the three major clades of bilaterians, what (if anything) can be inferred about whether the common ancestor of the clade had a coelom? (b) To what extent has the presence of a coelom in animals changed over the course of evolution?

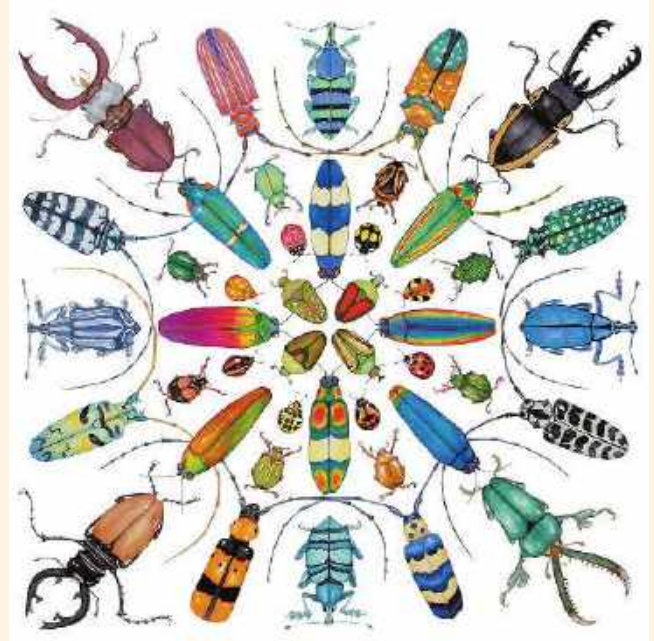
8. **SCIENTIFIC INQUIRY** Bats emit ultrasonic sounds and then use the returning echoes of those sounds to locate and capture flying insects, such as moths, in the dark. In response to bat attacks, some tiger moths make ultrasonic clicks of their own. Researchers hypothesize that tiger moth clicks likely either (1) jam the bat's sonar or (2) warn the bat about the moth's toxic chemical defenses. The graph below shows two patterns observed in studies of moth capture rates over time.



Bats in these experiments were “naive,” meaning that prior to the study the bats had not previously hunted tiger moths. Indicate whether the results support hypothesis (1), hypothesis (2), or both. Explain why the researchers used naive bats in this study.

9. **WRITE ABOUT A THEME: ORGANIZATION** Write a short essay (100–150 words) that explains how the structure of the digestive tract in different invertebrate groups affects the size of the organisms that they can eat.

10. SYNTHESIZE YOUR KNOWLEDGE



Collectively, do these beetles and all other invertebrate species combined form a monophyletic group? Explain your answer and provide an overview of the evolutionary history of invertebrate life.

For selected answers, see Appendix A.